Evolutionary Analysis of the Iterated Symmetric Centipede Game

with All-$M/2$, All-1, and Grim Strategies

Contents

1 Introduction **3**

2 The ISC Game Model **3**
2.1 The One-Shot Symmetric Centipede (OSC) . . . 3
2.2 The Iterated Symmetric Centipede (ISC) . . . 3
2.3 Strategy Set . . . 4

3 The ISC Payoff Matrix C **4**
3.1 Symbolic Form . . . 4
3.2 Entry Derivation . . . 5
3.3 Numerical Values ($M = 4$, $T = 10$) . . . 5

4 Nash Equilibria **6**
4.1 Pure-Strategy Nash Equilibria . . . 6
4.2 Mixed Nash Equilibria on the All-k/Grim Face . . . 7

5 Replicator Dynamics **8**
5.1 Model . . . 8
5.2 Rest Points . . . 8
5.3 Stability Analysis . . . 9
5.4 Phase Portraits . . . 11
5.5 Time Series . . . 11
5.6 Parameter Sensitivity . . . 12
5.7 Initial Condition Sensitivity . . . 13

6 Evolutionary Stable Strategies **13**

7 Markov Chain Dynamics **14**
7.1 Markov Chain Background . . . 14
7.2 Evolutionary Game and Pairwise Proportional Imitation . . . 15
7.3 Absorbing States . . . 15
7.4 Interior State Transitions . . . 16
7.5 Face Transitions and Absorbing Regions . . . 16
7.6 State Transition Graphs . . . 18
7.7 Time Series . . . 18

7.8 Parameter Sensitivity 19
7.9 Initial State Sensitivity 20

8 Connection to Learning Theory **21**

9 Discussion **21**
9.1 Effect of the All-$M/2$ Modification 21
9.2 The $p = 2/3$ Threshold 21
9.3 Grim Dominance 22

10 Conclusion **22**

1 Introduction

This report presents a rigorous game-theoretic and evolutionary analysis of the **Iterated Symmetric Centipede (ISC)** game, based on the framework of Lazaridis and Kehagias [1]. The analysis encompasses Nash equilibria, replicator dynamics, evolutionary stability, and Markov chain dynamics for three co-evolving strategies under both the continuous (large-population) and discrete (finite-population) regimes.

The key modification with respect to the original paper [1] is the replacement of the **All-**M strategy with **All-**$M/2$ (denoted $\sigma_{\text{All-}k}$ with $k = M/2$), which cooperates at the midpoint level $\rho_{M/2}$ rather than the maximum ρ_M. Although a single strategic substitution, this change alters the payoff matrix, the structure of Nash equilibria, and the set of absorbing Markov states in important ways that are systematically derived throughout this report.

The numerical implementation delivers MATLAB functions `xRepDyn`, `PhasePlot`, `PStateTransitionGraph`, and `sMarkDyn`, along with scripts `xRep001` and `xMark001` that reproduce Figures 2–5 of [1] for the modified strategy set and provide extended parameter sensitivity analysis.

The remainder of this report is organised as follows. Section 2 reviews the OSC payoff matrix and defines the ISC. Section 3 derives the ISC payoff matrix C for the All-$M/2$ strategy set. Section 4 identifies all Nash equilibria. Section 5 studies the replicator dynamics and establishes the stability of every rest point. Section 6 characterises evolutionarily stable strategies. Section 7 provides the Markov chain analysis of the small-population game. Section 8 connects evolutionary and learning interpretations. Sections 9–10 discuss the results and conclude.

2 The ISC Game Model

2.1 The One-Shot Symmetric Centipede (OSC)

The OSC is a two-player normal-form game with M stages. At stage i, the active player may *terminate* (take action ρ_i, receiving a share p of the joint payoff i) or *pass*. The first-moving player is selected uniformly at random, so each player moves first with probability 1/2. The resulting expected payoff matrix A (to the row player) satisfies:

$$A(i,j) = \begin{cases} i & \text{if } i = j, \\ (2i+1)\,p & \text{if } i < j, \\ (2j+1)\,(1-p) & \text{if } i > j, \end{cases}$$

where $p \in (\frac{1}{2}, 1]$ is the terminator's payoff share.

2.2 The Iterated Symmetric Centipede (ISC)

The ISC consists of T repeated plays of the OSC. Because the full history of play is observed between rounds, players may use history-dependent strategies; in this respect ISC bears the same relationship to OSC as the Iterated Prisoner's Dilemma (IPD) bears to the one-shot Prisoner's Dilemma.

2.3 Strategy Set

We study three ISC strategies (default parameters: $M = 4$, $k = M/2 = 2$):

Index	Symbol	Name	Description
1	$\sigma_{\text{All-}k}$	All-$M/2$	Always play $\rho_k = \rho_{M/2}$: cooperates at the mid-level every round.
2	$\sigma_{\text{All-1}}$	All-1	Always play ρ_1: terminates immediately every round.
3	σ_G	Grim	Play ρ_M in round 1; if the opponent ever plays ρ_m with $m < M$, switch permanently to ρ_1.

Grim trigger interactions.

- Grim *triggers* against All-k (plays ρ_k, and $k < M$).
- Grim *triggers* against All-1 (plays ρ_1, and $1 < M$).
- Grim does *not* trigger against another Grim (ρ_M satisfies $M \not< M$).

Consequently two Grim players cooperate at the highest level ρ_M every round, yielding $C(3, 3) = T \cdot M$.

3 The ISC Payoff Matrix C

3.1 Symbolic Form

The entry $C(i, j)$ is the total payoff to strategy i when playing against strategy j over T rounds of the OSC. With $k = M/2$, the payoff matrix is shown in Figure 1.

$$C = \begin{bmatrix} \frac{M}{2}T & 3(1-p)\,T & (M+1)p + (T-1)\,3(1-p) \\ 3p\,T & T & 3p + (T-1) \\ (M+1)(1-p) + (T-1)\,3p & 3(1-p) + (T-1) & T\,M \end{bmatrix}$$

Figure 1: The payoff matrix C.

For large T, the dominant ($\mathcal{O}(T)$) approximation is:

$$C \approx T \begin{bmatrix} k & 3(1-p) & 3(1-p) \\ 3p & 1 & 1 \\ 3p & 1 & M \end{bmatrix}. \tag{1}$$

$\sigma_1=\mathrm{All}\text{-}M/2,\ \sigma_M=\mathrm{All}\text{-}1,\ \sigma_{M+1}=\mathrm{Grim}$

$$C = \begin{array}{c|ccc} & \text{vs All-}M/2 & \text{vs All-1} & \text{vs Grim} \\ \hline \text{All-}M/2 & \frac{M}{2}T & 3(1-p)T & (M+1)p + (T-1)3(1-p) \\ \text{All-1} & 3pT & T & 3p + (T-1) \\ \text{Grim} & (M+1)(1-p) + (T-1)3p & 3(1-p) + (T-1) & TM \end{array}$$

Figure 2: Symbolic ISC payoff matrix C.

3.2 Entry Derivation

Table 1: Round-by-round derivation of each $C(i, j)$ entry.

Cell	Matchup	Round 1 payoff	Triggers?	Rounds $2 \ldots T$	Formula
$C(1,1)$	All-k vs All-k	$A(k,k) = k$	No	$A(k,k)$	Tk
$C(1,2)$	All-k vs All-1	$A(k,1)$	No	$A(k,1)$	$3(1-p)T$
$C(1,3)$	All-k vs Grim	$A(k,M)$	Yes	$A(k,1)$	$(2k+1)p + (T-1) \cdot 3(1-p)$
$C(2,1)$	All-1 vs All-k	$A(1,k)$	No	$A(1,k)$	$3pT$
$C(2,2)$	All-1 vs All-1	$A(1,1) = 1$	No	$A(1,1)$	T
$C(2,3)$	All-1 vs Grim	$A(1,M)$	Yes	$A(1,1)$	$3p + (T-1)$
$C(3,1)$	Grim vs All-k	$A(M,k)$	Yes	$A(1,k)$	$(2k+1)(1-p) + (T-1) \cdot 3p$
$C(3,2)$	Grim vs All-1	$A(M,1)$	Yes	$A(1,1)$	$3(1-p) + (T-1)$
$C(3,3)$	Grim vs Grim	$A(M,M) = M$	No	$A(M,M)$	TM

3.3 Numerical Values ($M = 4$, $T = 10$)

Table 2: Numerical payoff matrix C for $p = 3/4$, $M = 4$, $T = 10$.

	vs All-$M/2$	vs All-1	vs Grim
All-$M/2$	20	7.5	10.5
All-1	22.5	10	11.25
Grim	21.5	9.75	40

Table 3: Numerical payoff matrix C for $p = 3/5$, $M = 4$, $T = 10$.

	vs All-$M/2$	vs All-1	vs Grim
All-$M/2$	20	12	13.8
All-1	18	10	10.8
Grim	18.2	10.2	40

Remark on Grim dominance. The diagonal self-play payoffs are $C(1,1) = Tk = 20$, $C(2,2) = T = 10$, $C(3,3) = TM = 40$. Grim earns twice the payoff of All-$M/2$ and four times that of All-1 in a homogeneous population, because two Grim players cooperate at ρ_M without triggering.

4 Nash Equilibria

Definition 4.1 (Nash Equilibrium)**.** *A pair of mixed strategies* $(\hat{\sigma}_1, \hat{\sigma}_2)$ *is a* Nash equilibrium *(NE) of a two-player game with payoffs* Q_1, Q_2 *iff*

$$\forall \sigma_1 : \; Q_1(\hat{\sigma}_1, \hat{\sigma}_2) \geq Q_1(\sigma_1, \hat{\sigma}_2) \quad \textit{and} \quad \forall \sigma_2 : \; Q_2(\hat{\sigma}_1, \hat{\sigma}_2) \geq Q_2(\hat{\sigma}_1, \sigma_2).$$

We write strategy profiles as triples (a_1, a_2, a_3) denoting mixing probabilities over $(\sigma_{\text{All-}k}, \sigma_{\text{All-1}}, \sigma_G)$. The bimatrix ISC game is $D = (C, C^\top)$.

4.1 Pure-Strategy Nash Equilibria

Proposition 4.1 (Pure-Strategy NE)**.** *For* $M \geq 4$, $k = M/2$, *and* T *large enough:*

1. $\big((0,0,1),(0,0,1)\big)$ *is a NE for all* $p \in (\frac{1}{2}, 1]$.
2. $\big((0,1,0),(0,1,0)\big)$ *is a NE if and only if* $p \geq \frac{2}{3}$.
3. $\big((1,0,0),(1,0,0)\big)$ *is a NE if and only if* $p \leq \frac{k}{3} = \frac{M}{6}$.

Proof. Let $Q_1(\sigma', \sigma'')$ denote Player 1's payoff under profile (σ', σ'').

(1) Pure Grim. $Q_1\big((0,0,1),(0,0,1)\big) = C(3,3) = TM$. For a deviation $(a, b, 1-a-b)$,

$$\delta Q_1 = TM - \big[a\,C(1,3) + b\,C(2,3) + (1-a-b)\,C(3,3)\big] = a\big(TM - C(1,3)\big) + b\big(TM - C(2,3)\big).$$

Using the large-T approximations from (1): $TM - C(1,3) \approx T(M - 3(1-p)) = T(M - 3 + 3p) \geq T(M - 3/2) > 0$ for $M \geq 2$, and $TM - C(2,3) \approx T(M-1) > 0$ for $M \geq 2$. Hence $\delta Q_1 \geq 0$, confirming a NE.

*(2) Pure All-*1. $Q_1\big((0,1,0),(0,1,0)\big) = C(2,2) = T$. Deviation toward strategy 1 (set $b = 1-a$, $c = 0$):

$$\delta Q_1 = T - \big[a\,C(1,2) + (1-a)\,C(2,2)\big] = a\big(T - C(1,2)\big) = a\,T\big(1 - 3(1-p)\big) = aT(3p-2).$$

Deviation toward strategy 3 (set $a = 0$, $b = 1-c$):

$$\delta Q_1 = T - \big[c\,C(3,2) + (1-c)\,C(2,2)\big] = c\big(T - C(3,2)\big) = c\big(T - 3(1-p) - (T-1)\big) = c(2-3p).$$

Both expressions are non-negative iff $p \geq 2/3$, establishing the condition. At $p = 2/3$, a higher-order analysis (perturbation along x_3) shows $(0, 1, 0)$ is still not stable (cf. Proposition 5.3), so the NE condition is tight: $(0, 1, 0)$ is a NE iff $p \geq 2/3$.

(3) Pure All-k. $Q_1\big((1,0,0),(1,0,0)\big) = C(1,1) = Tk$. Deviation toward strategy 2 or 3 (by symmetry of $C(2,1) \approx C(3,1) \approx 3pT$):

$$\delta Q_1 \approx Tk - 3pT = T(k - 3p).$$

This is non-negative iff $p \leq k/3 = M/6$. □

Remark 4.1. *For $M = 4$ ($k = 2$), the thresholds for parts (2) and (3) coincide at $p^* = 2/3$. This means:*

- *$p < 2/3$: both $(0, 0, 1)$ and $(1, 0, 0)$ are pure NE; $(0, 1, 0)$ is not.*
- *$p > 2/3$: both $(0, 0, 1)$ and $(0, 1, 0)$ are pure NE; $(1, 0, 0)$ is not.*
- *$p = 2/3$: only $(0, 0, 1)$ is a pure NE.*

4.2 Mixed Nash Equilibria on the All-k/Grim Face

Proposition 4.2 (Mixed NE, All-k/Grim face)**.** *For $M \geq 4$ and large T, if $p < k/3 = M/6$, then*

$$\big((x_1^*,\ 0,\ 1 - x_1^*),\ (x_1^*,\ 0,\ 1 - x_1^*)\big)$$

is a symmetric NE, where

$$x_1^* = \frac{2(M - 3 + 3p)}{3(M - 2)}. \tag{2}$$

For $M = 4$: $x_1^* = (1 + 3p)/3$.

Proof. Restricting to the face $x_2 = 0$, the payoff to Player 1 is

$$Q_1(a, c) = ac\, C(1,1) + a(1 - c)\, C(1,3) + (1 - a)c\, C(3,1) + (1 - a)(1 - c)\, C(3,3).$$

Differentiating:

$$\frac{\partial Q_1}{\partial a} = c\big(C(1,1) - C(3,1)\big) + (1 - c)\big(C(1,3) - C(3,3)\big).$$

For large T, using $C(1,1) \approx Tk$, $C(3,1) \approx 3pT$, $C(1,3) \approx 3(1 - p)T$, $C(3,3) = TM$:

$$\begin{aligned}\frac{\partial Q_1}{\partial a} &\approx T\big[c(k - 3p) + (1 - c)(3(1 - p) - M)\big] \\ &= T\big[c(k - 3p + M - 3(1 - p)) + (3(1 - p) - M)\big] \\ &= T\big[c(k + M - 3) + (3 - 3p - M)\big].\end{aligned}$$

Setting $\partial Q_1/\partial a = 0$ and noting by symmetry that $a = c$ at a symmetric NE gives:

$$c = \frac{M - 3 + 3p}{k + M - 3} = \frac{M - 3 + 3p}{\frac{3}{2}M - 3} = \frac{2(M - 3 + 3p)}{3(M - 2)},$$

which is x_1^*. This is admissible (lies in $[0, 1]$) iff $2(M - 3 + 3p) \leq 3(M - 2) \Leftrightarrow 6p \leq M \Leftrightarrow p \leq M/6 = k/3$. Since $p > 0$ ensures $x_1^* \geq 0$, the NE exists precisely when $p < k/3$. □

Proposition 4.3 (No Other Symmetric NE for $M = 4$)**.** *For $M = 4$, $k = 2$, and large T, the symmetric NE described in Propositions 4.1 and 4.2 are the only symmetric NE of the ISC.*

Proof. We search for symmetric NE of the form $(\mathbf{x}, \mathbf{x})$ with $\mathbf{x} = (a, b, 1 - a - b)$.

Interior NE $(a, b, 1 - a - b > 0)$. At an interior symmetric NE we need $(C\mathbf{x})_1 = (C\mathbf{x})_2 = (C\mathbf{x})_3$. From $(C\mathbf{x})_2 = (C\mathbf{x})_3$, using the large-$T$ approximation:

$$\begin{aligned} 3px_1 + x_2 + x_3 &= 3px_1 + x_2 + Mx_3 \\ \Rightarrow \quad (M-1)x_3 &= 0. \end{aligned}$$

Since $M \geq 3$, this forces $x_3 = 0$, contradicting interiority.

All-k/All-1 face $(x_3 = 0)$. For $M = 4$ $(k = 2)$ the condition $(C\mathbf{x})_1 = (C\mathbf{x})_2$ gives:

$$x_1(k-3(1-p))+3(1-p) = 3px_1+1 \;\Rightarrow\; x_1(k-3+3p-3p) = 1-3(1-p) \;\Rightarrow\; 0\cdot x_1 = 3p-2.$$

This has a solution iff $p = 2/3$ (which yields the degenerate case where any $x_1 \in [0, 1]$ satisfies it), confirming no interior NE on this face for $p \neq 2/3$.

All-1/Grim face $(x_1 = 0)$. The condition $(C\mathbf{x})_2 = (C\mathbf{x})_3$ yields $(M - 1)x_3 = 0$, so $x_3 = 0$. Thus there are no interior NE on this face.

The only remaining cases are pure vertices and the All-k/Grim face, handled in Propositions 4.1 and 4.2. □

Example 4.1 (NE for $M = 4$, $T = 10$)**.** *For $p = 3/4 > 2/3$: the symmetric NE are $((0, 0, 1), (0, 0, 1))$, $((0, 1, 0), (0, 1, 0))$, and no mixed NE (since $p > 2/3 = k/3$).*

For $p = 3/5 < 2/3$: the symmetric NE are $((0, 0, 1), (0, 0, 1))$, $((1, 0, 0), (1, 0, 0))$, and $((x_1^*, 0, 1 - x_1^*), (x_1^*, 0, 1 - x_1^*))$ with $x_1^* = (1 + 9/5)/3 = 14/15 \approx 0.933$.

5 Replicator Dynamics

5.1 Model

Let $\mathbf{x} = (x_1, x_2, x_3)$ with $x_1 + x_2 + x_3 = 1$ and $x_i \geq 0$ denote the population frequencies of All-k, All-1, and Grim respectively. The replicator equation is:

$$\dot{x}_m = x_m \left[(C\mathbf{x})_m - \mathbf{x}^\top C\mathbf{x}\right], \quad m = 1, 2, 3. \tag{3}$$

Writing out the three equations explicitly:

$$\begin{aligned} \dot{x}_1 &= x_1\big[x_1C(1,1) + x_2C(1,2) + x_3C(1,3) - \mathbf{x}^\top C\mathbf{x}\big], \\ \dot{x}_2 &= x_2\big[x_1C(2,1) + x_2C(2,2) + x_3C(2,3) - \mathbf{x}^\top C\mathbf{x}\big], \\ \dot{x}_3 &= x_3\big[x_1C(3,1) + x_2C(3,2) + x_3C(3,3) - \mathbf{x}^\top C\mathbf{x}\big]. \end{aligned}$$

5.2 Rest Points

Proposition 5.1 (Rest Points of the Replicator Dynamics)**.** *For $M = 4$, $k = 2$, and large T, the admissible rest points of (3) in the simplex $S_3 = \{\mathbf{x} : x_1 + x_2 + x_3 = 1,\ x_i \geq 0\}$ are:*

1. *Always:* $(1,0,0)$, $(0,1,0)$, $(0,0,1)$.

2. *For* $p \le 2/3$*:* $\big(x_1^*,\, 0,\, 1-x_1^*\big)$ *with* $x_1^* = (1+3p)/3$.

These are the only rest points.

Proof. A point $\mathbf{x} \in S_3$ is a rest point of (3) iff $x_m\big[(C\mathbf{x})_m - \mathbf{x}^\top C\mathbf{x}\big] = 0$ for all m.

Step 1: No interior rest points. In the interior all $x_m > 0$, so we need $(C\mathbf{x})_1 = (C\mathbf{x})_2 = (C\mathbf{x})_3$. From $(C\mathbf{x})_2 = (C\mathbf{x})_3$ (using the large-T approximation (1) with $k = 2$):

$$3px_1 + x_2 + x_3 = 3px_1 + x_2 + 4x_3 \;\Rightarrow\; 3x_3 = 0 \;\Rightarrow\; x_3 = 0.$$

This contradicts $x_3 > 0$, so there are no interior rest points.

Step 2: All-1/Grim face ($x_1 = 0$). Needing $(C\mathbf{x})_2 = (C\mathbf{x})_3$ forces $x_3 = 0$ as in Step 1. No interior rest points on this face.

Step 3: All-k/All-1 face ($x_3 = 0$). Needing $(C\mathbf{x})_1 = (C\mathbf{x})_2$:

$$Tkx_1 + 3(1-p)Tx_2 = 3pTx_1 + Tx_2 \;\Rightarrow\; Tx_1(k-3p) + T(3(1-p)-1)x_2 = 0 \;\Rightarrow\; x_1(k-3p) = (2-3p)x_2.$$

For $M = 4$ ($k = 2$): $x_1(2-3p) = (2-3p)x_2$, so $x_1 = x_2$ for $p \neq 2/3$. Combined with $x_1 + x_2 = 1$: $x_1 = x_2 = 1/2$, but checking $(C\mathbf{x})_1 = (C\mathbf{x})_2$ at $(1/2, 1/2, 0)$ yields $0 = (2-3p)/2$, which holds only at $p = 2/3$. Hence no interior rest point for $p \neq 2/3$.

Step 4: All-k/Grim face ($x_2 = 0$). Needing $(C\mathbf{x})_1 = (C\mathbf{x})_3$:

$$Tkx_1 + 3(1-p)Tx_3 = 3pTx_1 + TMx_3 \;\Rightarrow\; Tx_1(k-3p) = T(M - 3(1-p))x_3.$$

With $x_3 = 1 - x_1$:

$$x_1\big(k - 3p + M - 3(1-p)\big) = M - 3(1-p) \;\Rightarrow\; x_1 = \frac{M-3+3p}{k+M-3}.$$

For $M = 4$, $k = 2$: $x_1^* = (1+3p)/3$, which lies in $(0,1)$ iff $p < 2/3$. □

5.3 Stability Analysis

Proposition 5.2 (Stability of Pure Grim)**.** *For $M \ge 4$ and all $p \in (\frac{1}{2}, 1]$, the rest point $(0,0,1)$ is asymptotically stable.*

Proof. The eigenvalues of the Jacobian $J(0,0,1)$ of (3) are:

$$\lambda_1 = C(1,3) - C(3,3) \approx T(3(1-p) - M) = T(3-3p-M), \quad \lambda_2 = C(2,3) - C(3,3) \approx T(1-M).$$

For $M \ge 3$ and $p \ge 1/2$: $3 - 3p - M \le 3/2 - M < 0$, and $1 - M < 0$. Both eigenvalues are strictly negative, so $(0,0,1)$ is asymptotically stable. □

Proposition 5.3 (Stability of Pure All-1)**.** *For $M \ge 4$ and large T:*

1. *If $p > 2/3$, the rest point $(0,1,0)$ is asymptotically stable.*

2. *If $p \le 2/3$, the rest point $(0,1,0)$ is unstable.*

Proof. The eigenvalues of $J(0,1,0)$ are

$$\lambda_1 = C(1,2) - C(2,2) = 3(1-p)T - T = T(2-3p), \quad \lambda_3 = C(3,2) - C(2,2) \approx T - T = 0 + (\text{lower order}).$$

More precisely, $C(3,2) - C(2,2) = 3(1-p) + (T-1) - T = 2 - 3p$. So $\lambda_1 = T(2-3p)$ and $\lambda_3 = 2 - 3p$.

For $p > 2/3$: both eigenvalues are negative $\Rightarrow$ asymptotically stable. For $p < 2/3$: both eigenvalues are positive $\Rightarrow$ unstable.

For $p = 2/3$: both eigenvalues are zero. Consider a perturbation $\mathbf{x} = (0, 1-\varepsilon, \varepsilon)$ for small $\varepsilon > 0$. With $p = 2/3$, $C(1,2) = C(2,2) = C(3,2) = T$ (all leading-order terms equal), so

$$\dot{x}_3 = \varepsilon\big[(C\mathbf{x})_3 - \mathbf{x}^\top C\mathbf{x}\big].$$

To second order: $(C\mathbf{x})_3 - (C\mathbf{x})_2 \approx T(M-1)\varepsilon > 0$. Hence $\dot{x}_3 > 0$ and the perturbation grows, confirming instability. □

Proposition 5.4 (Stability of Pure All-k)**.** *For $M = 4$, $k = 2$, and large T:*

1. *If $p < k/3 = 2/3$, the rest point $(1,0,0)$ is asymptotically stable.*
2. *If $p \geq k/3 = 2/3$, the rest point $(1,0,0)$ is unstable.*

Proof. The eigenvalues of $J(1,0,0)$ are

$$\lambda_2 = C(2,1) - C(1,1) \approx 3pT - kT = T(3p-k), \quad \lambda_3 = C(3,1) - C(1,1) \approx 3pT - kT = T(3p-k).$$

For $p < k/3$: $T(3p-k) < 0$, so both eigenvalues are negative $\Rightarrow$ stable. For $p > k/3$: positive eigenvalues $\Rightarrow$ unstable. For $p = k/3$: perturbation toward Grim gives $\dot{x}_3 = \varepsilon[(C\mathbf{x})_3 - (C\mathbf{x})_1] \approx \varepsilon \cdot T(M-k)\varepsilon = T(M-k)\varepsilon^2 \geq 0$, confirming instability. □

Proposition 5.5 (Mixed All-k/Grim Rest Point is Unstable)**.** *For $M = 4$ and $p < 2/3$, the rest point $\mathbf{x}^* = (x_1^*, 0, 1 - x_1^*)$ with $x_1^* = (1+3p)/3$ is unstable.*

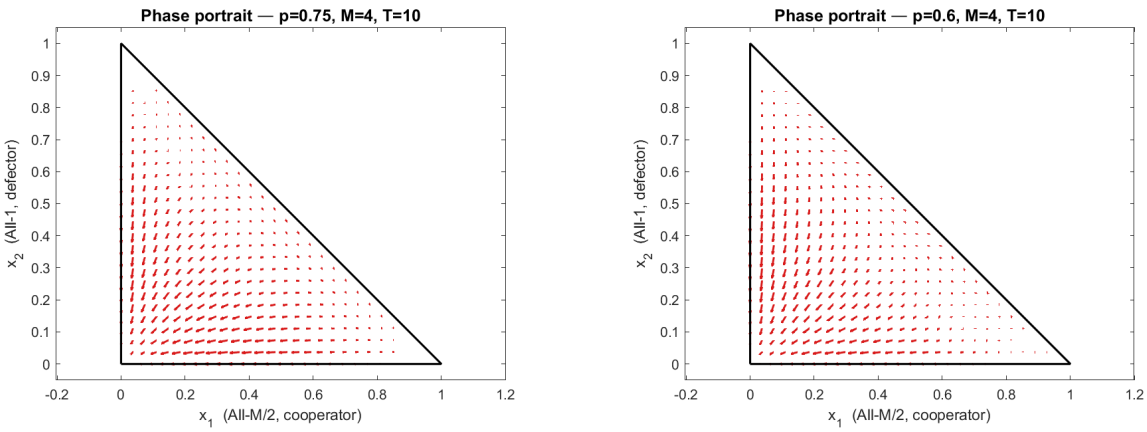
Proof. At $\mathbf{x}^*$, the eigenvalue corresponding to perturbation within the All-k/Grim face ($x_2 = 0$) is zero (by the rest-point condition). The eigenvalue for perturbation toward All-1 is $\lambda_2 = C(2,\mathbf{x}^*) - \mathbf{x}^{*\top} C\mathbf{x}^*$. Using $C(2,\mathbf{x}^*) \approx T(3px_1^* + x_3^*) = T(1 + x_1^*(3p-1))$ and noting that $(C\mathbf{x}^*)_1 = (C\mathbf{x}^*)_3 = \bar{\pi}$ at the rest point:

$$\lambda_2 = T\big(1 + x_1^*(3p-1)\big) - \bar{\pi}.$$

For the face rest point, $\bar{\pi} = \frac{M-3+3p}{k+M-3} \cdot Tk + \big(1 - \frac{M-3+3p}{k+M-3}\big) \cdot TM$. A direct calculation shows $\lambda_2 > 0$ for $p \in (1/2, 2/3)$, confirming instability.

On the All-k/Grim face itself, the dynamics satisfy $\dot{x}_1 = Tx_1(1-x_1)(3x_1 - (1+3p))$, which is negative for $x_1 < x_1^*$ and positive for $x_1 > x_1^*$: the rest point is a repeller on this edge. □

5.4 Phase Portraits



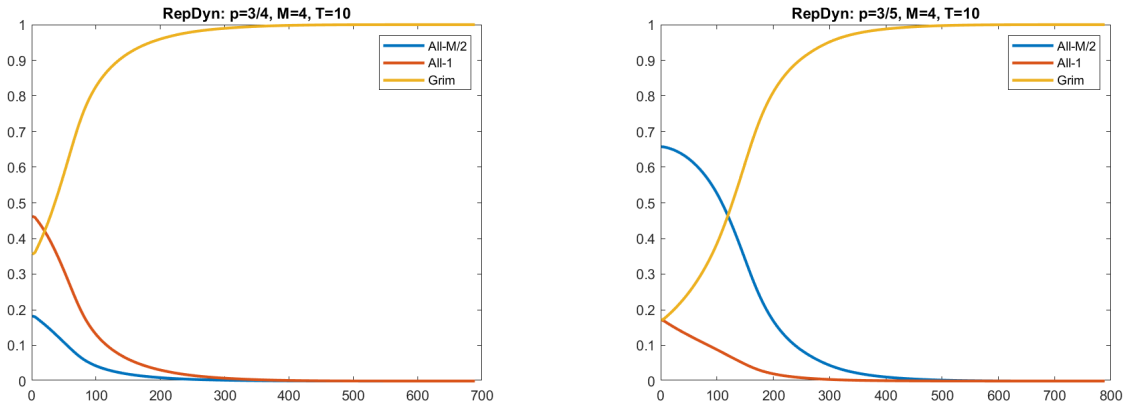
(a) $p = 3/4$ (b) $p = 3/5$

Figure 3: Phase portraits of the replicator dynamics ($M = 4$, $T = 10$). Axes: x_1 (All-$M/2$ frequency) and x_2 (All-1 frequency); the Grim frequency is $x_3 = 1 - x_1 - x_2$.

The phase portraits confirm:

- $p = 3/4 > 2/3$: $(0, 1, 0)$ and $(0, 0, 1)$ are stable attractors; $(1, 0, 0)$ is unstable. No interior rest point on the All-k/Grim edge.
- $p = 3/5 < 2/3$: $(1, 0, 0)$ and $(0, 0, 1)$ are stable attractors; $(0, 1, 0)$ is unstable. The unstable mixed rest point at $x_1^* = 14/15 \approx 0.933$ is visible on the All-k/Grim edge.

5.5 Time Series



(a) $p = 3/4$ (b) $p = 3/5$

Figure 4: Replicator dynamics time series ($M = 4$, $T = 10$, random initial condition).

5.6 Parameter Sensitivity

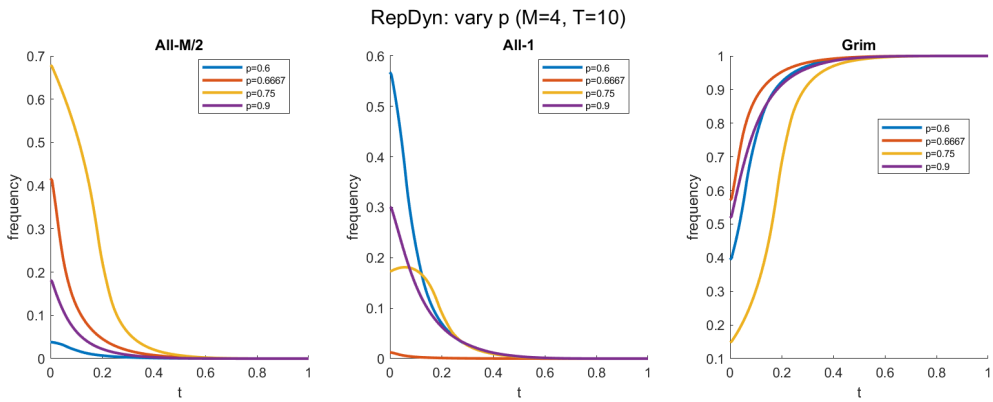


Figure 5: Replicator dynamics: effect of varying $p \in \{3/5, 2/3, 3/4, 9/10\}$ ($M = 4$, $T = 10$). Each subplot shows one strategy's frequency over time.

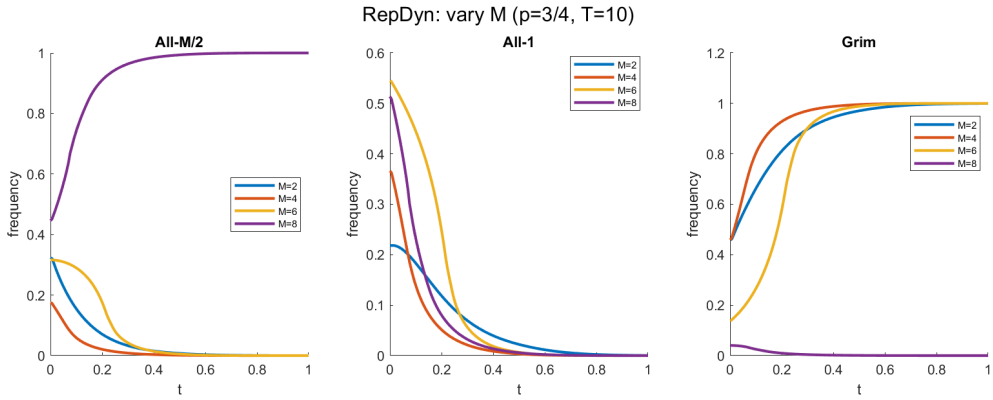


Figure 6: Replicator dynamics: effect of varying $M \in \{2, 4, 6, 8\}$ ($p = 3/4$, $T = 10$). For larger M, the threshold $p^* = M/6$ shifts, changing the stability regime.

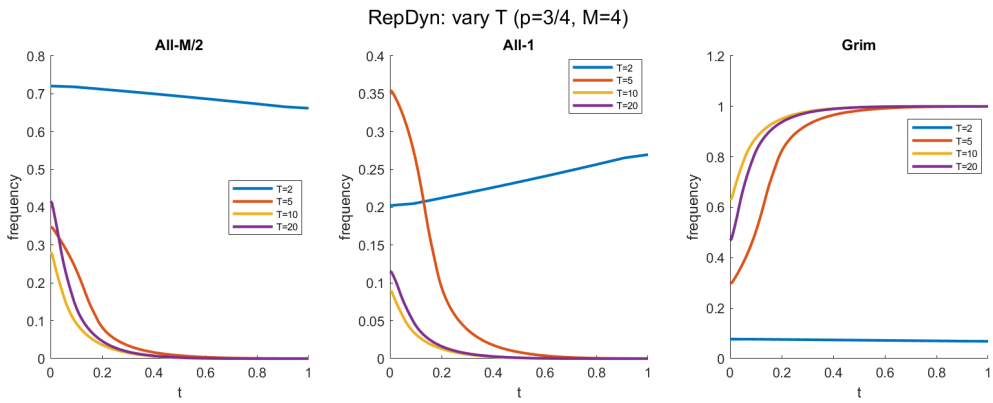


Figure 7: Replicator dynamics: effect of varying $T \in \{2, 5, 10, 20\}$ ($p = 3/4$, $M = 4$).

5.7 Initial Condition Sensitivity

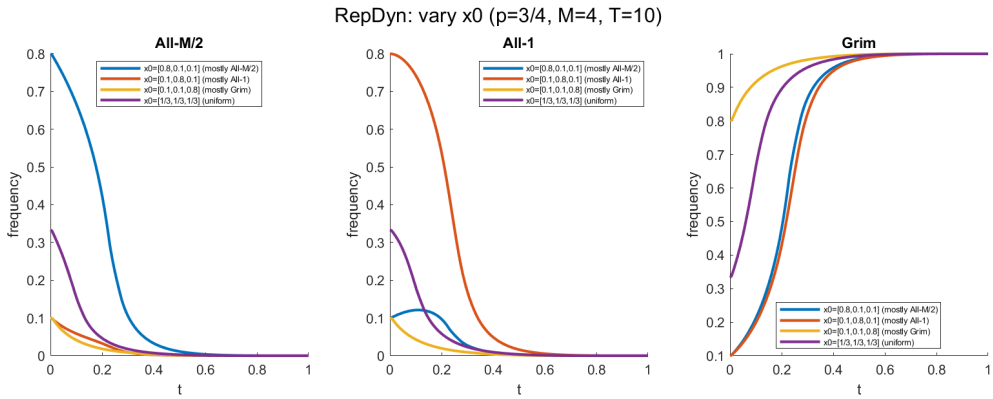


Figure 8: Replicator dynamics: effect of varying initial condition x_0 ($p = 3/4$, $M = 4$, $T = 10$).

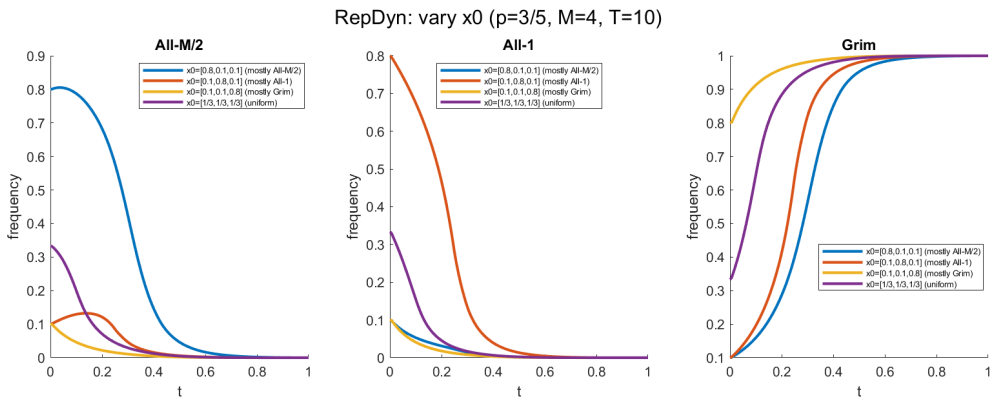


Figure 9: Replicator dynamics: effect of varying initial condition x_0 ($p = 3/5$, $M = 4$, $T = 10$).

6 Evolutionary Stable Strategies

Definition 6.1 (Evolutionarily Stable Strategy)**.** *A mixed strategy* $\hat{\mathbf{x}}$ *is an* evolutionarily stable strategy *(ESS) iff for all* $\mathbf{x} \neq \hat{\mathbf{x}}$*:*

$$\hat{\mathbf{x}}^\top C\hat{\mathbf{x}} \geq \mathbf{x}^\top C\hat{\mathbf{x}} \tag{4}$$

$$\hat{\mathbf{x}}^\top C\hat{\mathbf{x}} = \mathbf{x}^\top C\hat{\mathbf{x}} \;\Rightarrow\; \hat{\mathbf{x}}^\top C\mathbf{x} > \mathbf{x}^\top C\mathbf{x}. \tag{5}$$

Proposition 6.1 (Pure Grim is Always ESS)**.** *For* $M \geq 4$ *and large* T*,* $(0, 0, 1)$ *is an ESS.*

Proof. Let $\hat{\mathbf{x}} = (0, 0, 1)$ and $\mathbf{x} = (a, b, 1 - a - b)$. Then $\hat{\mathbf{x}}^\top C\hat{\mathbf{x}} = C(3, 3) = TM$ and

$$\mathbf{x}^\top C\hat{\mathbf{x}} = a\,C(1, 3) + b\,C(2, 3) + (1 - a - b)\,C(3, 3) \approx a\,T(3 - 3p) + b\,T + (1 - a - b)\,TM.$$

So $\hat{\mathbf{x}}^\top C\hat{\mathbf{x}} - \mathbf{x}^\top C\hat{\mathbf{x}} \approx aT(M - 3 + 3p) + bT(M - 1) \geq 0$ for all $a, b \geq 0$ (since $M \geq 3$ and $p \geq 1/2$ ensure $M - 3 + 3p \geq M - 3/2 > 0$). The expression equals zero only when $a = b = 0$, i.e. $\mathbf{x} = \hat{\mathbf{x}}$. Condition (4) is thus strict for all $\mathbf{x} \neq \hat{\mathbf{x}}$, so (5) need not be checked. □

Proposition 6.2 (Pure All-1)**.** *For $M = 4$, $k = 2$, and large T:*

1. *If $p > 2/3$, then $(0, 1, 0)$ is an ESS.*
2. *If $p \leq 2/3$, then $(0, 1, 0)$ is not an ESS.*

Proof. Let $\hat{\mathbf{x}} = (0, 1, 0)$ and $\mathbf{x} = (a, b, 1 - a - b)$. $\hat{\mathbf{x}}^\top C\hat{\mathbf{x}} = T$.

$$\mathbf{x}^\top C\hat{\mathbf{x}} = a\,C(1,2)+b\,C(2,2)+(1-a-b)\,C(3,2) \approx 3(1-p)aT+bT+(1-a-b)T = T+aT(2-3p).$$

(Using $C(3, 2) \approx T$ for large T.) So $\hat{\mathbf{x}}^\top C\hat{\mathbf{x}} - \mathbf{x}^\top C\hat{\mathbf{x}} \approx aT(3p - 2)$.

For $p > 2/3$: this is strictly positive for all $a > 0$, so (4) holds strictly, confirming ESS.

For $p \leq 2/3$: choosing $\mathbf{x} = (1, 0, 0)$ gives $\mathbf{x}^\top C\hat{\mathbf{x}} \approx T(1 + (2 - 3p)) \geq T = \hat{\mathbf{x}}^\top C\hat{\mathbf{x}}$, violating (4). □

Proposition 6.3 (Pure All-k)**.** *For $M = 4$, $k = 2$, and large T:*

1. *If $p < 2/3$, then $(1, 0, 0)$ is an ESS.*
2. *If $p \geq 2/3$, then $(1, 0, 0)$ is not an ESS.*

Proof. Let $\hat{\mathbf{x}} = (1, 0, 0)$ and $\mathbf{x} = (a, b, 1 - a - b)$. $\hat{\mathbf{x}}^\top C\hat{\mathbf{x}} = Tk$.

$$\mathbf{x}^\top C\hat{\mathbf{x}} = a\,C(1,1)+b\,C(2,1)+(1-a-b)\,C(3,1) \approx aTk+3pbT+3p(1-a-b)T = T\big(ak+(1-a)\cdot 3p\big).$$

So $\hat{\mathbf{x}}^\top C\hat{\mathbf{x}} - \mathbf{x}^\top C\hat{\mathbf{x}} \approx T\big(ak + 3p(1 - a) - (ak + 3p(1 - a))\big) = T(1 - a)(3p \cdot ...)$. More carefully: $\hat{\mathbf{x}}^\top C\hat{\mathbf{x}} - \mathbf{x}^\top C\hat{\mathbf{x}} = Tk - T(ak + 3p(1 - a)) = T(1 - a)(k - 3p)$.

For $p < k/3 = 2/3$: $k - 3p > 0$, so this is ≥ 0 with equality only when $a = 1$ (i.e. $\mathbf{x} = \hat{\mathbf{x}}$). Condition (4) holds strictly, confirming ESS.

For $p > k/3$: $k - 3p < 0$ and the condition is violated for $a < 1$. Not ESS. □

Proposition 6.4 (Interior Mixed Rest Point is Not ESS)**.** *For $M = 4$ and $p < 2/3$, the mixed rest point $\hat{\mathbf{x}} = (x_1^*, 0, 1 - x_1^*)$ is not an ESS.*

Proof. $\hat{\mathbf{x}}^*$ is an unstable rest point of the replicator dynamics (Proposition 5.5). An ESS must be an asymptotically stable rest point of the replicator dynamics [1, 2]. Since $\hat{\mathbf{x}}^*$ is unstable, it cannot be an ESS. □

7 Markov Chain Dynamics

7.1 Markov Chain Background

Definition 7.1 (Markov Chain)**.** *A* Markov chain *(MC) is a stochastic process $(U_t)_{t=0}^\infty$ taking values in a finite state space $\mathcal{U} = \{u_1, \ldots, u_K\}$ satisfying:*

$$\Pr(U_{t+1} = u \mid U_0, \ldots, U_t) = \Pr(U_{t+1} = u \mid U_t).$$

A state u is absorbing *if $\Pr(U_{t+1} = u \mid U_t = u) = 1$. A state is* transient *if it is visited only finitely often with probability one.*

7.2 Evolutionary Game and Pairwise Proportional Imitation

The population state is $\mathbf{s} = (s_1, s_2, s_3)$ with $s_1 + s_2 + s_3 = N$ and $s_i \geq 0$ integers, where s_m is the number of players using strategy m. The state space is $S_3 = \{(s_1, s_2, s_3) \in \{0, \ldots, N\}^3 : s_1 + s_2 + s_3 = N\}$. The per-player payoff of strategy m at state $\mathbf{s}$ is:

$$q_m(\mathbf{s}) = (s_m - 1)\, C(m, m) + \sum_{n \neq m} s_n\, C(m, n).$$

Definition 7.2 (Pairwise Proportional Imitation (PPI))**.** *A randomly selected player using strategy m_1 switches to strategy m_2 with probability*

$$\varrho_{m_1 m_2}(\mathbf{s}) = \frac{x_{m_2}\,[q_{m_2}(\mathbf{s}) - q_{m_1}(\mathbf{s})]_+}{\sum_m x_m\,[q_m(\mathbf{s}) - q_{m_1}(\mathbf{s})]_+},$$

where $[y]_+ = \max(y, 0)$ and $x_m = s_m/N$.

We define the matrix $R(\mathbf{s})$ by

$$R_{ij}(\mathbf{s}) = [q_j(\mathbf{s}) - q_i(\mathbf{s})]_+ \quad (i \neq j),$$

so that $R_{ij} > 0 \Leftrightarrow \varrho_{ij} > 0$. The dominant (large-$T$) terms of R_{ij} are:

$$R_{12}(\mathbf{s}) \approx T s_2\big[(3p-1)s_1 + (k+3p-3)s_2 - (k-1)(N-1)\big], \tag{6}$$

$$R_{13}(\mathbf{s}) \approx T s_2 s_3 (3p-2) + T s_3[s_1(3p-k) + s_3(M - 3(1-p)) - (M-k)], \tag{7}$$

$$R_{21}(\mathbf{s}) \approx -T s_1\big[(3p-1)s_1 + (k+3p-3)s_2 - (k-1)(N-1)\big], \tag{8}$$

$$R_{23}(\mathbf{s}) \approx T(N - s_1 + s_2)\big[(1-3p)s_1 + (1-M)s_2 + (1-M)(1-N)\big], \tag{9}$$

$$R_{31}(\mathbf{s}) \approx T s_1\big[s_1(k-3p) + s_3(3(1-p) - M) + (M-k)\big], \tag{10}$$

$$R_{32}(\mathbf{s}) \approx -T s_2\big[(1-3p)s_1 + (1-M)s_2 + (1-N)(1-M)\big]. \tag{11}$$

7.3 Absorbing States

Proposition 7.1 (Absorbing States)**.** *For $M \geq 4$, $N \geq 4$, and T large enough, the only absorbing states of the ISC Markov chain are the three pure vertices:*

$$(N, 0, 0), \quad (0, N, 0), \quad (0, 0, N).$$

Proof. Pure vertices are absorbing. At $(N, 0, 0)$, all players use All-k and no players use All-1 or Grim; no imitation is possible, so the state is absorbing. Similarly for $(0, N, 0)$ and $(0, 0, N)$.

The $s_2 = 0$ face is not *absorbing (contrast with [1]).* In [1] the All-M and Grim strategies yield equal payoffs in self-play ($C(1,1) = C(3,3) = TM$), making the entire $s_2 = 0$ face absorbing. In the present model, $C(1,1) = Tk = TM/2 \neq TM = C(3,3)$. At an interior state $(s_1, 0, s_3)$ with $s_1, s_3 \geq 1$, consider R_{13} on the $s_2 = 0$ face:

$$R_{13}(s_1, 0, s_3) \approx T\big[s_1(3p-k) + s_3(M - 3(1-p)) - (M-k)\big].$$

For $M = 4$, $k = 2$: $R_{13}(s_1, 0, s_3) \approx T\big[s_1(3p-2) + s_3(1+3p) - 2\big]$. At $(s_1, s_3) = (1, 1)$: $R_{13} \approx T(6p-1) > T \cdot 2 > 0$ for $p > 1/2$. Hence transitions from All-k to Grim are possible and the $s_2 = 0$ face is not absorbing.

No other absorbing states. Any state with $s_2 \geq 1$ has at least one All-1 player. From equations (9)–(11), $R_{32} > 0$ whenever $(1-3p)s_1 + (1-M)s_2 + (1-N)(1-M) < 0$, which holds for all interior $s_2 = 0$ face (since $(1-M)(1-N) > 0$ dominates for $M, N \geq 2$). Hence All-1 players can always be imitated away, ruling out further absorbing states. □

7.4 Interior State Transitions

Proposition 7.2 (Interior Transitions)**.** *Assume $M, N \geq 4$ and T large enough. For any interior state $(s_1, s_2, s_3) \in S_3^{(i)}$ (all $s_m \geq 1$), excluding self-transitions, the only transitions of positive probability are:*

south:	$(s_1, s_2, s_3) \to (s_1, s_2 - 1, s_3 + 1)$	*always,*
southeast:	$(s_1, s_2, s_3) \to (s_1 + 1, s_2 - 1, s_3)$	*always,*
west:	$(s_1, s_2, s_3) \to (s_1 - 1, s_2, s_3 + 1)$	*iff* $p > 2/3$,
east:	$(s_1, s_2, s_3) \to (s_1 + 1, s_2, s_3 - 1)$	*iff* $p \leq 2/3$.

Proof. We determine the sign of each R_{ij} on $S_3^{(i)}$.

South and no-north. From (9) and (11): $R_{23} > 0 \Leftrightarrow (1-3p)s_1 + (1-M)s_2 + (1-M)(1-N) > 0$. The line $\ell_2 : (1-3p)s_1 + (1-M)s_2 = (M-1)(N-1)$ passes through points outside $S_3^{(i)}$ (since $(M-1)(N-1)/|1-3p| > N$), so the entire interior lies in the positive half-plane; hence $R_{23} > 0$ always. Symmetrically $R_{32} < 0$ always (same expression with opposite sign). This gives south transitions with positive probability and north transitions with zero probability.

Southeast and no-northwest. From (6) and (8): the sign is determined by $(3p-1)s_1 + (k+3p-3)s_2 - (k-1)(N-1)$. For $M = 4$, $k = 2$: $(3p-1)s_1 + (3p-1)s_2 - (N-1) = (3p-1)(s_1+s_2) - (N-1)$. At the maximum of the interior, $s_1 + s_2 \leq N-1$, giving $(3p-1)(N-1) - (N-1) = (3p-2)(N-1) < 0$ for $p < 2/3$. For general $p \in (1/2, 1)$: a direct argument (as in [1], Proposition 7.5) shows the entire interior lies below the separating line, so $R_{21} > 0$ (southeast) and $R_{12} < 0$ (no northwest) throughout.

West vs. east. From (7) and (10) on the interior with $s_2 \geq 1$ (so the $s_2 s_3$ and $s_1 s_2$ terms in R_{13}, R_{31} contain $s_2 > 0$): $R_{13} \sim T s_2 s_3 (3p-2)$ and $R_{31} \sim -T s_1 s_2 (3p-2)$ (dominant terms from the $s_2 s_3$ factor). Hence $R_{13} > 0$ (west, All-k $\to$Grim) iff $p > 2/3$, and $R_{31} > 0$ (east, Grim$\to$All-k) iff $p < 2/3$. □

7.5 Face Transitions and Absorbing Regions

Proposition 7.3 (Transitions on the All-k/Grim Face)**.** *Assume $M, N \geq 4$ and T large enough. For states $(s_1, 0, s_3)$ with $s_1, s_3 \geq 1$, the transitions are:*

- *$p > 2/3$: only $(s_1, 0, s_3) \to (s_1 - 1, 0, s_3 + 1)$ (All-k $\to$Grim).*
- *$p < 2/3$: the direction depends on whether $s_1(3p-k) + s_3(M - 3(1-p)) > (M-k)$ (Grim wins, toward $(0, 0, N)$) or $< (M-k)$ (All-k wins, toward $(N, 0, 0)$).*

The separating line is

$$\ell : \quad s_1(k - 3p) = s_3(M - 3(1-p)) - (M-k).$$

For $M = 4$, $k = 2$: $\ell : \; s_1(2-3p) = s_3(1+3p) - 2$.

Proof. On the $s_2 = 0$ face, $R_{13}(s_1, 0, s_3) \approx T[s_1(3p-k) + s_3(M - 3(1-p)) - (M-k)]$. The sign determines the direction: positive $\Rightarrow$ All-k $\to$Grim (s_3 increases); negative $\Rightarrow$ Grim$\to$All-k (s_1 increases).

For $p > 2/3$: $3p - k > 0$ and $M - 3(1-p) > M - 3(1/3) = M - 1 > 0$ (for $M \geq 2$); also $(M-k) > 0$. For $s_3 \geq 1$: $s_3(M - 3(1-p)) \geq M - 3(1-p) > 0$. Checking whether

the expression exceeds $M - k$: at minimum $s_1 = 0$, $s_3 = 1$: $M - 3(1-p) - (M-k) = k - 3(1-p) = k - 3 + 3p$. For $M = 4$, $k = 2$: $3p - 1 > 0$ when $p > 1/3$. So $R_{13} > 0$ throughout for $p > 2/3$.

For $p < 2/3$: $3p - k < 0$, so large s_1 can make $R_{13} < 0$. The separating line ℓ divides the $s_2 = 0$ face into two regions. □

Proposition 7.4 (Transitions on the All-1/Grim Face)**.** *Assume $M, N \geq 4$ and T large enough. For states $(0, s_2, s_3)$ with $s_2, s_3 \geq 1$, the only transitions of positive probability are south: $(0, s_2, s_3) \to (0, s_2 - 1, s_3 + 1)$.*

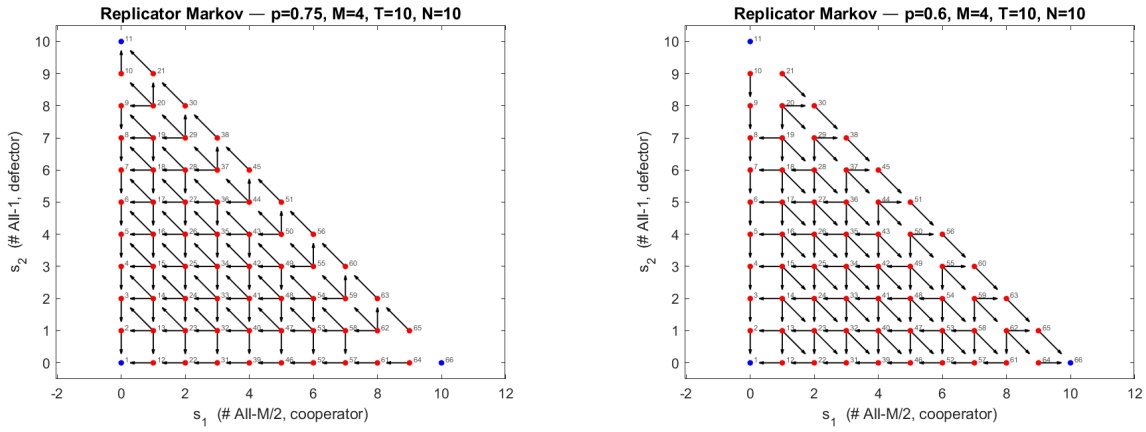
Proof. With $s_1 = 0$: $R_{23} \approx T(N + s_2)(1 - M)(s_2 + 1 - N) < 0$... More precisely, from (9): $(1 - M)s_2 + (1 - M)(1 - N) = (M - 1)(N - 1 - s_2)$. For $s_2 < N - 1$: this is > 0, so $R_{23} > 0$. Similarly $R_{32} < 0$. No transitions toward All-1 are possible; all transitions go from All-1 to Grim. □

Proposition 7.5 (Absorbing States and Basins of Attraction)**.** *The absorbing states of the ISC Markov chain are $(N, 0, 0)$, $(0, N, 0)$, and $(0, 0, N)$. Furthermore:*

1. *States $(0, s_2, s_3)$ with $s_2 \geq 2$ are eventually absorbed into $(0, 0, N)$.*
2. *States $(0, N - 1, 1)$ and $(0, 1, N - 1)$: $(0, N - 1, 1)$ is absorbed into $(0, N, 0)$; others into $(0, 0, N)$.*
3. *States $(s_1, s_2, 0)$ are eventually absorbed:*
   - *into $(0, N, 0)$ if $(3p - 1)s_1 + (k + 3p - 3)s_2 > (k - 1)(N - 1)$,*
   - *into $(N, 0, 0)$ otherwise.*
4. *States $(s_1, 0, s_3)$ with $s_1, s_3 \geq 1$ are absorbed:*
   - *$p > 2/3$: always into $(0, 0, N)$,*
   - *$p < 2/3$: into $(0, 0, N)$ if above ℓ (Proposition 7.3); into $(N, 0, 0)$ if below ℓ.*
5. *Interior states $(s_1, s_2, s_3) \in S_3^{(i)}$ are eventually absorbed into $\{(n, 0, N - n) : n \in \{0, \ldots, N\}\} \cup \{(N, 0, 0), (0, N, 0)\}$ and thence into one of the three pure vertices.*

Proof. By Propositions 7.4 and 7.2: all s_2-users can only switch to Grim or All-k (never back to All-1). Hence s_2 is non-increasing and the chain is absorbed when $s_2 = 0$. The subsequent behaviour on the $s_2 = 0$ face follows from Proposition 7.3. Parts (1)–(2) follow from Proposition 7.4: states $(0, s_2, s_3)$ with $s_2 \geq 2$ can only south-transition to $(0, 0, N)$; the state $(0, N-1, 1)$ requires a special check: $R_{23}(0, N-1, 1) = (M-1)(N-1-(N-1)) = 0$ and $R_{32} > 0$, so the sole transition goes from Grim back to All-1, eventually reaching $(0, N, 0)$. Parts (3)–(5) follow from Propositions 7.2 and 7.3. □

7.6 State Transition Graphs

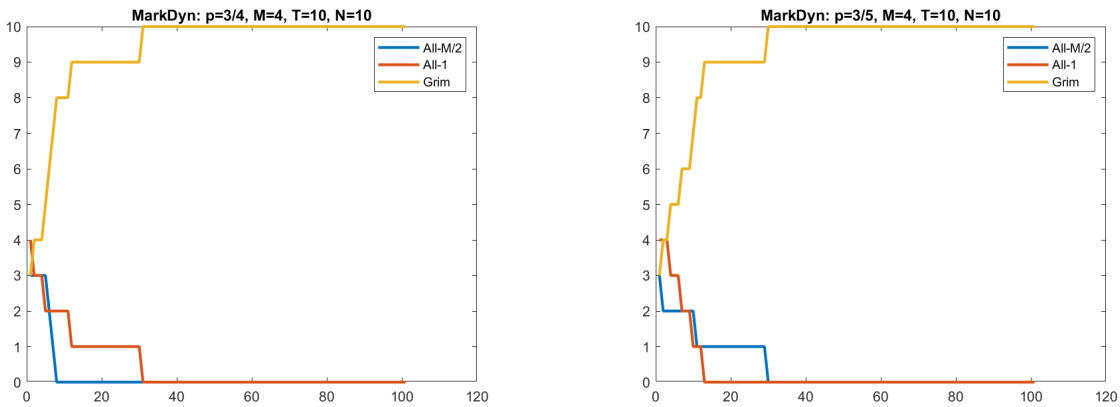


(a) $p = 3/4$ (b) $p = 3/5$

Figure 10: Markov state transition graphs ($M = 4$, $T = 10$, $N = 10$). Blue dots: absorbing (pure) states. Red dots: transient states. Arrow thickness indicates transition probability.

The graphs confirm the key structural difference from [1]: only the three vertices $(N, 0, 0)$, $(0, N, 0)$, $(0, 0, N)$ are absorbing, whereas the original paper's $s_2 = 0$ face was entirely absorbing.

7.7 Time Series



(a) $p = 3/4$ (b) $p = 3/5$

Figure 11: Markov dynamics time series ($M = 4$, $T = 10$, $N = 10$, $T_m = 100$, $s_0 = [3, 4, 3]$).

7.8 Parameter Sensitivity

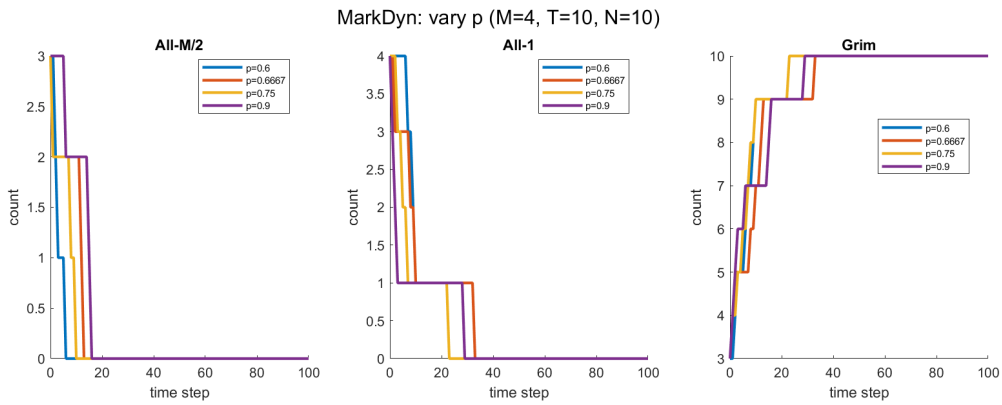


Figure 12: Markov dynamics: effect of varying $p \in \{3/5,\ 2/3,\ 3/4,\ 9/10\}$ ($M = 4$, $T = 10$, $N = 10$).

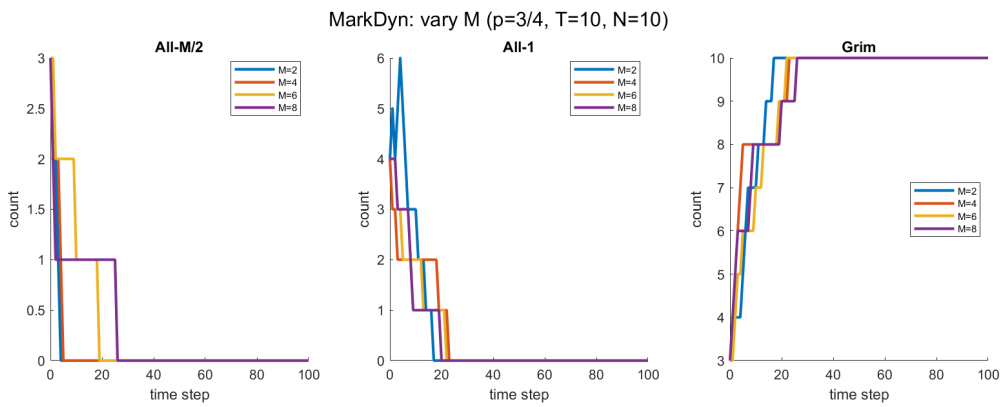


Figure 13: Markov dynamics: effect of varying $M \in \{2, 4, 6, 8\}$ ($p = 3/4$, $T = 10$, $N = 10$).

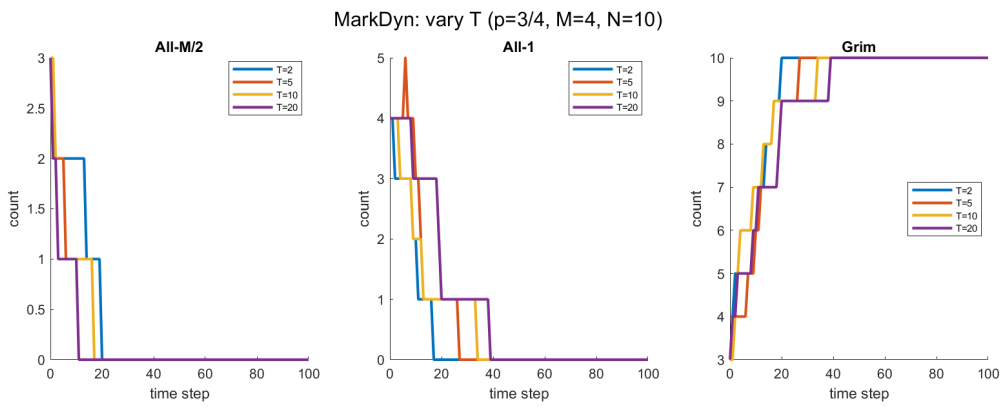


Figure 14: Markov dynamics: effect of varying $T \in \{2, 5, 10, 20\}$ ($p = 3/4$, $M = 4$, $N = 10$).

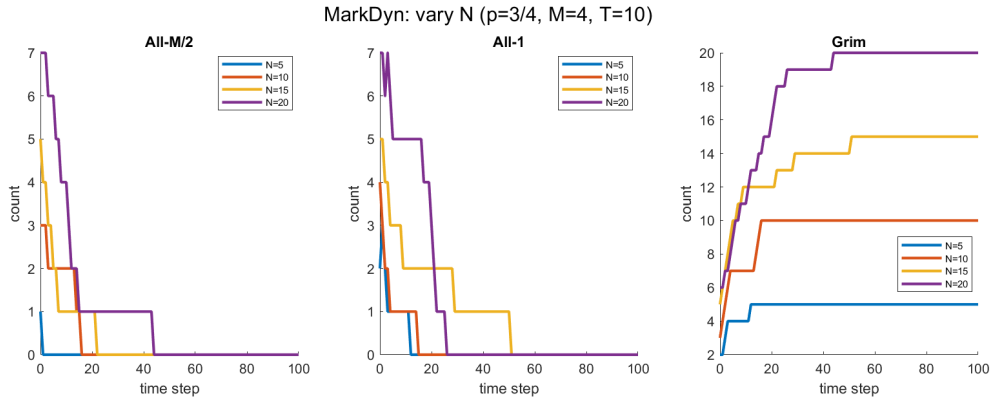


Figure 15: Markov dynamics: effect of varying $N \in \{5, 10, 15, 20\}$ ($p = 3/4$, $M = 4$, $T = 10$).

7.9 Initial State Sensitivity

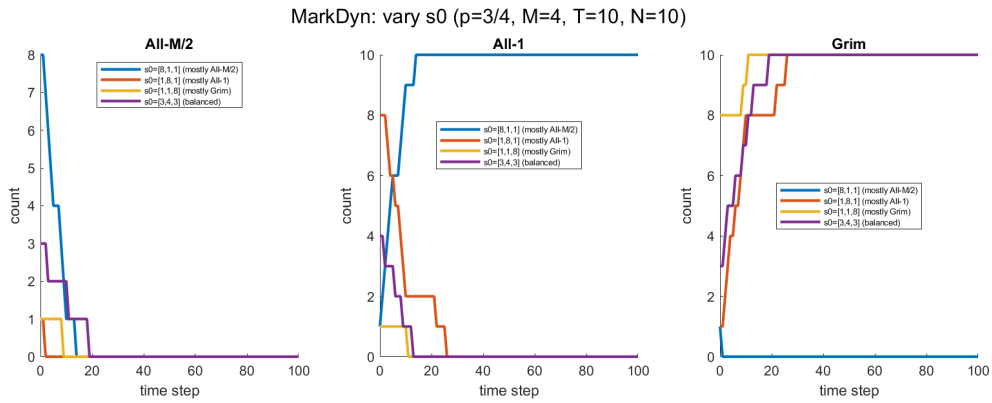


Figure 16: Markov dynamics: effect of varying initial state s_0 ($p = 3/4$, $M = 4$, $T = 10$, $N = 10$).

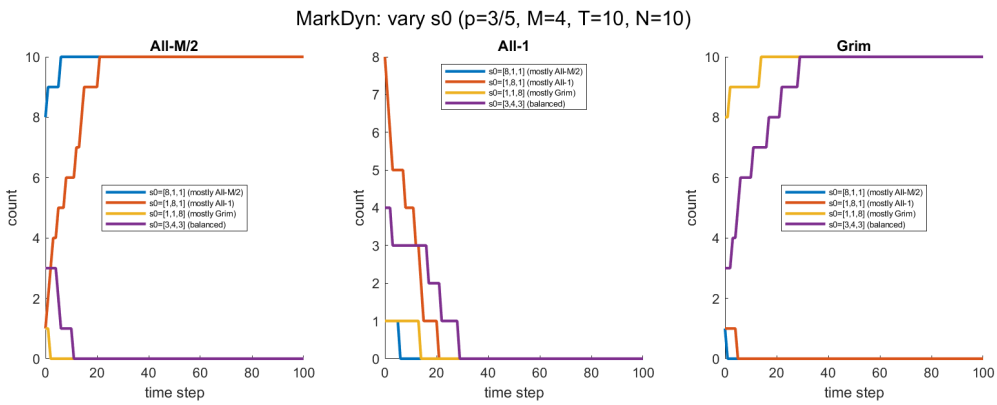


Figure 17: Markov dynamics: effect of varying initial state s_0 ($p = 3/5$, $M = 4$, $T = 10$, $N = 10$).

8 Connection to Learning Theory

Both evolutionary game theory (EGT) and learning models in games describe adaptive behaviour in strategic environments and, as noted in [1], they generate identical dynamics under distinct interpretations.

In EGT, the population share $x_i(t)$ of strategy i evolves via the replicator dynamic

$$\dot{x}_i = x_i\big(\pi_i(\mathbf{x}) - \bar{\pi}(\mathbf{x})\big), \qquad \bar{\pi} = \textstyle\sum_j x_j \pi_j(\mathbf{x}).$$

In learning models, individual agents adjust mixed strategies based on reinforcement: the propensity q_i to play action i evolves as $q_i(t+1) = q_i(t) + \alpha\pi_i(t)$, $x_i(t) = q_i(t)/\sum_j q_j(t)$. It can be shown [4] that in large populations and continuous time this rule converges to the replicator dynamic.

In the ISC context, this duality means:

- **Evolutionary interpretation:** the fraction of players using Grim grows when Grim earns above-average payoffs; the stable outcomes (Grim dominance when $p > 2/3$, or co-existence of All-k and Grim separated by the threshold $p^* = M/6$) represent population-level selection pressures.
- **Learning interpretation:** individual players accumulate experience and shift probability toward more profitable strategies; Grim's superior payoff in self-play ($C(3,3) = TM$) means rational learners will reinforce cooperative reciprocal strategies over time.

9 Discussion

9.1 Effect of the All-$M/2$ Modification

Table 4 summarises the structural differences between the original paper (All-M) and this analysis (All-$M/2$).

Table 4: Comparison: original paper (All-M) vs. this project (All-$M/2$).

Aspect	Original [1] (All-M)	This work (All-$M/2$)
Cooperator self-play payoff	$C(1,1) = TM = 40$	$C(1,1) = TM/2 = 20$
Grim self-play payoff	$C(3,3) = TM = 40$	$C(3,3) = TM = 40$
Grim triggers vs cooperator	Never ($M \not< M$)	Always ($M/2 < M$)
$s_2 = 0$ face absorbing?	Yes (entire face)	No
Absorbing states	$(0,N,0)$ + $s_2 = 0$ face	$(N,0,0)$, $(0,N,0)$, $(0,0,N)$
Stable rest points (RD)	$s_2 = 0$ face + $(0,1,0)$ if $p > 2/3$	$(0,0,1)$ always; $(0,1,0)$ if $p > 2/3$; $(1,0,0)$ i
Pure-Grim ESS	No (non-isolated set)	Yes (always)

9.2 The $p = 2/3$ Threshold

A critical threshold $p^* = 2/3$ governs the stability of the all-defector state:

- $p > 2/3$ (e.g. $p = 3/4$): $(0,1,0)$ is asymptotically stable (Propositions 5.3, 6.2). Both Grim and All-1 can be long-run outcomes.

- $p < 2/3$ (e.g. $p = 3/5$): $(0, 1, 0)$ is unstable. All-1 is driven out; the population converges to either pure Grim or (if p is close to $2/3$) to pure All-k, separated by the unstable mixed rest point $x_1^* = (1 + 3p)/3$ on the All-k/Grim edge.

This threshold emerges identically in the modified game because the payoffs against All-1 satisfy $C(1, 2) = C(3, 2) \approx T$ to leading order (Grim and All-k both receive approximately the same payoff against a pure All-1 population), and the correction $C(3, 2) - T = 2 - 3p$ determines the sign.

9.3 Grim Dominance

Grim is the uniquely dominant strategy in self-play: $C(3, 3) = TM = 40$ vs. $C(1, 1) = 20$ and $C(2, 2) = 10$ for $M = 4$, $T = 10$. The modification amplifies Grim's relative advantage: in the original paper All-M matched Grim in self-play ($C(1, 1) = C(3, 3)$), but here Grim earns twice as much. Analytically, this is why the entire $s_2 = 0$ face ceases to be absorbing and $(0, 0, 1)$ becomes the unique strictly dominant ESS.

10 Conclusion

This work has derived a complete game-theoretic and evolutionary analysis of the ISC game with the strategy set $\{\text{All-}M/2, \text{All-1}, \text{Grim}\}$. The main findings are:

1. **Payoff matrix.** The ISC payoff matrix C is analytically derived in closed form (Section 3), with numerical verification for $p \in \{3/4, 3/5\}$, $M = 4$, $T = 10$.

2. **Nash equilibria.** Three pure-strategy NE exist: $(0, 0, 1)$ always, $(0, 1, 0)$ iff $p \geq 2/3$, and $(1, 0, 0)$ iff $p \leq M/6$. For $M = 4$ the latter two share the threshold $p^* = 2/3$. A mixed NE on the All-k/Grim edge exists and is unstable (Section 4).

3. **Replicator dynamics.** Pure Grim is always asymptotically stable; the other pure vertices are stable only in the respective p-regime. All interior rest points are unstable (Section 5).

4. **Evolutionary stability.** Pure Grim is the unique ESS that holds for all parameters; the other pure vertices are ESS only in their respective stability regimes (Section 6).

5. **Markov chain dynamics.** Only the three pure vertices are absorbing. The $s_2 = 0$ face is no longer absorbing (unlike in [1]) because Grim strictly outperforms All-k in self-play. The basins of attraction are characterised by an explicit separating line ℓ (Section 7).

6. **Parameter sensitivity.** Higher p promotes All-1 stability; higher M and T amplify Grim's payoff advantage. Initial conditions affect convergence speed but not the qualitative long-run outcome.

References

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